AROGYAMITRA HEALTHCARE SYSTEM

DOCTOR ADDITION WORKFLOW & DATABASE SCHEMA

Comprehensive Backend & Frontend Technical Specification

This document provides a clear, end-to-end technical explanation of how an Administrator adds doctors to a hospital in ArogyaMitra, how doctor profiles are linked to hospitals via Foreign Keys, and the exact database table schema used in the PostgreSQL backend.

1. WHICH TABLE STORES DOCTORS IN BACKEND?

All doctors across all hospitals are stored in a centralized master table named:

```
public.doctors (PostgreSQL Master Table)
```

Key relationship: Each doctor has a foreign key link to the hospital table:

```
FOREIGN KEY (hospital_id) REFERENCES public.hospitals(id) ON DELETE CASCADE
```

Detailed Column Schema for `public.doctors` Table:

Column Name	Data Type	Constraint	Purpose / Description
id	UUID	PRIMARY KEY	Unique identifier (gen_random_uuid())
hospital_id	UUID	FK -> hospitals(id)	Links doctor to specific hospital
doctor_code	VARCHAR(100)	OPTIONAL	Doctor staff ID code (e.g. DOC-9123)
name	VARCHAR(150)	NOT NULL	Full name of the doctor
registration_number	VARCHAR(100)	NULLABLE	State Medical Council Reg. No.
qualification	VARCHAR(255)	NULLABLE	Degrees (e.g. MBBS, MD, MS, DNB)
specialization	VARCHAR(255)	NULLABLE	Primary Speciality (e.g. Cardiology)
specialties	TEXT[] (Array)	NULLABLE	Multiple selectable clinical specialties
experience / _years	VARCHAR / INT	DEFAULT 0	Total clinical practice experience
doctor_type	VARCHAR(50)	DEFAULT 'Both'	OPD, IPD, or Both (OPD/IPD)
duty_type	VARCHAR(50)	DEFAULT 'Shift'	Shift / Full-Time / On-Call
working_days	TEXT[] (Array)	DEFAULT Mon-Sat	Assigned duty days in weekly roster
start_time / end_time	VARCHAR(20)	DEFAULT 04:00	Shift timing hours (e.g. 09:00 - 05:00)
opd_shift / opd_timing	VARCHAR	DEFAULT Morning	Display shift label & timing string
availability_status	VARCHAR(50)	DEFAULT 'Available'	Available, On Leave, In Surgery, Off Duty
contact_number	VARCHAR(30)	NULLABLE	Mobile number for urgent coordination
address / district / taluka	TEXT / VARCHAR	NULLABLE	Location details of the doctor
is_active / status	BOOLEAN / VARCHAR	DEFAULT TRUE / 'ACTIVE'	Active roster participation flag

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HOW ADMIN ADDS DOCTORS (STEP-BY-STEP WORKFLOW)

Frontend UI Action Flow & Backend Execution Pipeline

2. HOW ADMIN ADDS DOCTOR IN HOSPITAL (TWO SIMPLE WAYS)

In the ArogyaMitra system, an Admin can add doctors to a hospital through two simple methods:

METHOD A: During Hospital Creation or Hospital Edit (Recommended for Bulk Setup)

1. Admin navigates to Admin Portal -> Hospital Management -> "+ Add New Hospital" or clicks "Edit Hospital".

2. In Step 3 ("Doctors & Medical Staff"), Admin clicks "+ Add Doctor" button.

3. Admin fills doctor details: Name, Mobile, Specialities (multi-select), OPD/IPD Type, Shift Timings.

4. When Admin clicks "Submit Hospital", frontend sends payload with `doctors: [...]` array to backend.

5. Backend `hospital.service.js` creates the hospital first, gets `hospital.id`, and automatically inserts all doctors into `public.doctors` table with that `hospital_id`.

METHOD B: Independent Direct Addition via Doctor Management / Roster Modal

1. Admin opens Hospital Detail page -> Doctors Tab, or opens Doctor Management page.

2. Clicks "+ Add Doctor" / "Register Doctor" button to open the modal.

3. Selects Hospital from dropdown (or auto-filled with current hospital), fills doctor details.

4. Submits form -> Frontend calls `POST /api/doctors` with `hospital_id` in request body.

5. Backend `doctor.service.js` immediately inserts the doctor record directly into `public.doctors` table.

3. COMPLETE REST API REFERENCE FOR DOCTOR MANAGEMENT

HTTP Method & Route	Access Role	Service Function	Description
POST /api/doctors	Admin / Sevak	doctorService.createDoctor	Registers a new doctor under a hospital
GET /api/doctors	Public / Authenticated	doctorService.getAllDoctors	Lists doctors (filter by hospital_id, etc.)
GET /api/doctors/:id	Public / Authenticated	doctorService.getDoctorById	Fetches complete profile & roster of doctor
PUT /api/doctors/:id	Admin / Sevak	doctorService.updateDoctor	Updates doctor details, shift & specialties
PATCH /api/doctors/:id/availability	Admin / Sevak	doctorService.updateAvailability	1-Click toggle: Available, On Leave, Busy
DELETE /api/doctors/:id	Admin Only	doctorService.deleteDoctor	Removes doctor from hospital records

System Summary & Key Architecture Note:

All doctor records are stored in `public.doctors`. Deleting or deactivating a hospital automatically cascades or archives related doctors via PostgreSQL foreign key constraints.